Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_13

1. Summary of Key Deviations

The analysis of Run_13 reveals a critical failure in one engine's fuel feed, with the rest of the system showing minor, secondary effects.

- **Flow_Engine1 (Critical):** This is the primary and most severe anomaly. The signal for engine 1 shows a **major negative drift**, starting at -60.5% and culminating in a complete **signal dropout (-99.7%)** by state s5. This indicates a total loss of flow to one specific engine nozzle.
- **Flow_Bypass (Moderate):** A moderate **negative offset** (up to **-13.9%**) is observed. This is an unusual secondary symptom.
- **p_Combustion (Minor):** A consistent, minor **negative offset** of approximately **-6% to -7%** is present across all active states.
- **Flow_Engine2, 3, 4 (Minor):** The other three engine flows show a slight **positive offset** (up to +5.4%) during the high-throttle phases. This suggests a control system response attempting to compensate for the fault.
- **Flow_Pump & Pressures (Minor/Nominal):** `Flow_Pump` shows a negligible negative drift, and all other system pressures are either nominal or show very minor negative offsets. `Flow_PRV` is healthy.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Injector Nozzle 1**	**Severe Blockage / Clogging**	This is the most direct explanation for the **critical dropout of Flow_Engine1**. A complete blockage at this nozzle would stop flow to engine 1. The control system, sensing a drop in total fuel delivery or pressure, would likely command other components to compensate, explaining the slight positive offset on the other three engine flows. The negative bypass flow is unusual but could be a complex secondary effect of the redistribution of pressures and flows.

2	**Fuel Metering Unit (FMU)**	**Internal Blockage (Channel to Engine 1)**	A blockage or failure of the specific internal channel feeding Engine 1 within the FMU would produce symptoms identical to a clogged nozzle. It is an equally strong candidate for the physical location of the blockage.
3	**Flow_Engine1 Sensor**	**Sensor Dropout / Failure**	A sensor-only failure is possible. However, the compensatory increase in flow to the other three engines suggests the control system is reacting to a *real* physical problem. If the `Flow_Engine1` sensor had simply failed, it's less likely we would see a corresponding physical reaction from the other engine flows.
4	**Piping to Engine 1**	**Line Crimp / Blockage**	The physical fuel line running from the FMU to injector nozzle 1 could be crimped or blocked, which is physically equivalent to the other blockage hypotheses.
5	**System Contamination**	**Debris causing blockage**	This is the likely underlying *cause* of the blockage (a piece of debris). The main diagnostic challenge is pinpointing the *location* of this blockage, which is clearly in the Engine 1 fuel path.

3. Most Probable Hypothesis

The most probable root cause is a **Severe Blockage in the fuel path to Engine 1**, located either at the **Injector Nozzle 1** or within the corresponding **FMU channel**.

Reasoning: The evidence for a severe blockage in the Engine 1 line is definitive. The signal for `Flow_Engine1` drops to zero, while the other three engine signals remain stable or even increase slightly, indicating a highly localized fault. The system's attempt to compensate by increasing flow to the other engines is strong evidence that this is a real physical event, not just a sensor failure. The negative pressure and bypass flow readings are likely complex secondary effects of this primary blockage. The fault is clearly isolated to the individual fuel path for Engine 1.